BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION MO/2023)

CLASS: **BTECH** **SEMESTER : V**
BRANCH: **ECE/EEE/CSE** **SESSION : MO/2023**

SUBJECT: CS263 DATA STRUCTURES AND ALGORITHM

TIME: **02 Hours** **FULL MARKS: 25**

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
***4*. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates**

			CO	BL
Q.1(a)	Differentiate between linear and non-linear data structure	[2]	1	2
Q.1(b)	Define an algorithm. Explain its features with the help of suitable examples	[3]	1	1
Q.2(a)	Explain polynomial representation using link list	[2]	1	2
Q.2(b)	Define Big “oh” (O), Big Omega (Ω) and Big Theta (Θ) notations of time complexity.	[3]	1	1
Q.3(a)	Explain the advantages of link list over array?	[2]	3	2
Q.3(b)	Write a program to create a queue using arrays which permits insertion at both the ends	[3]	2	3
Q.4(a)	Describe sparse matrix with example?	[2]	1	1
Q.4(b)	Write a program to implement a stack that stores Character data.	[3]	2	4
Q.5(a)	Compare between a circular linked list and singly linked list	[2]	2	2
Q.5(b)	Convert the following infix expressions to their postfix equivalents (1). (A-B) + C*D/ E-C (2) (A- 2*(B+C)/D*E)+F	[3]	2	5

:::::27/09/2023 M:::::